Story2Proposal: A Scaffold for Structured Scientific Paper Writing

Story2Proposal Agent [1]

This manuscript, including its writing and figures, was primarily generated by the Story2Proposal Agent. Human contributors focused on designing and implementing the agent system, as well as collecting and organizing experimental results, without direct involvement in the writing process. As such, any imperfections in writing reflect the current capabilities of the system rather than extensive manual polishing.

Abstract

Generating scientific manuscripts requires maintaining alignment between narrative reasoning, experimental evidence, and visual artifacts across the document lifecycle. Existing language-model generation pipelines rely on unconstrained text synthesis with validation applied only after generation, often producing structural drift, missing figures or tables, and cross-section inconsistencies. We introduce Story2Proposal, a contract-governed multi-agent framework that converts a research story into a structured manuscript through coordinated agents operating under a persistent shared visual contract. The system organizes architect, writer, refiner, and renderer agents around a contract state that tracks section structure and registered visual elements, while evaluation agents supply feedback in a generate–evaluate–adapt loop that updates the contract during generation. Experiments on tasks derived from the Jericho research corpus show that Story2Proposal achieved an expert evaluation score of 6.145 versus 3.963 for DirectChat (+2.182) across GPT, Claude, Gemini, and Qwen backbones. Compared with the structured generation baseline Fars, Story2Proposal obtained an average score of 5.705 versus 5.197, indicating improved structural consistency and visual alignment.

1. Introduction

Academic paper generation represents a fundamental challenge in long-form text synthesis [1, 2, 3, 4, 5], where maintaining global coherence, argumentative consistency, and structural integrity across thousands of words remains difficult for current language models. Consider a researcher attempting to generate a complete conference paper from a high-level research concept: the introduction must establish motivation and contributions, the method section must provide technical details that align with those contributions, the experiments must validate the exact claims stated earlier, and the related work must position the approach against cited baselines without contradicting the methodology [6, 7, 8, 9, 10]. This requirement for cross-section consistency, claim-evidence alignment, and citation grounding distinguishes academic writing from other long-form generation tasks. Modular approaches to complex reasoning [11, 12, 13] and controllable generation frameworks [14, 15, 16] have demonstrated the value of intermediate representations for maintaining coherence, yet existing systems fail to provide the semantic control mechanisms necessary for paper-length document generation.

Current approaches to automated academic writing suffer from fundamental limitations in maintaining consistency across paper-length documents. Direct prompting methods [17, 18, 19, 20, 21] generate papers end-to-end without intermediate control mechanisms, leading to section drift where later sections diverge from the initial problem framing, claim-experiment misalignment where stated contributions lack corresponding empirical validation, citation-argument disconnection where references fail to ground specific technical claims, and inter-section contradictions where methodology descriptions conflict with experimental setups. Single-agent long-form generation systems [22, 23, 24, 25, 26] lack the specialized expertise needed for different paper sections, producing generic content that fails to capture domain-specific argumentation patterns. Outline-based approaches [27, 28, 29, 30, 31] provide high-level structure but offer insufficient semantic grounding to maintain argumentative coherence across sections. Agent frameworks with planning capabilities [32, 33, 34] demonstrate coordination mechanisms but do not address the provenance tracking required

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Preprint. March 23, 2026.

to ensure generated text remains faithful to source concepts. Controllable generation methods [35, 36, 37, 38, 39] establish the need for intermediate control but do not provide the structured semantic layer necessary for academic writing, where each claim must trace back to explicit research contributions and each experiment must validate stated hypotheses.

To address these limitations, we propose Story2Proposal, a contract-governed multi-agent framework for automated scientific manuscript generation that integrates collaborative agents with persistent structural governance. Instead of treating manuscript production as unconstrained text synthesis, Story2Proposal models it as a contract-governed construction process in which structural and visual obligations are explicitly represented and continuously enforced through a persistent shared visual contract. The framework coordinates four specialized agents: an architect agent that transforms a research story into a structured blueprint and initializes the contract state, a writer agent that generates section drafts under contract constraints, a refiner agent that improves coherence and global narrative alignment, and a renderer agent that materializes figures, tables, and LaTeX structure consistent with the contract. Throughout generation, the contract state records section structure, registered visual artifacts, and validation rules. Evaluation agents analyze intermediate outputs for reasoning quality, data fidelity, and visual consistency, producing feedback that updates the contract state and guides subsequent generation steps within a generate–evaluate–adapt loop.

We evaluate the framework across multiple large language model settings and manuscript-generation scenarios derived from the Jericho research corpus. Comparative experiments against strong baselines show that contract-governed generation improves structural consistency, visual integration, and overall manuscript coherence. The results indicate that persistent structural contracts stabilize multi-stage generation and reduce structural drift during iterative rewriting. Qualitative analysis further shows that the persistent shared visual contract preserves figure placement and maintains alignment between narrative reasoning and supporting artifacts during the document lifecycle.

Our main contributions are:

- We introduce Story2Proposal, a contract-governed multi-agent framework that coordinates architect, writer, refiner, and renderer agents through a persistent shared visual contract to maintain structural and visual consistency during scientific manuscript generation. Unlike DirectChat, which performs single-stage prompt-based generation without persistent constraints, Story2Proposal maintains a contract state that explicitly tracks section obligations and visual artifacts.
- We design a generate–evaluate–adapt mechanism that integrates evaluation agents for reasoning validation, data fidelity checking, and visual coherence assessment. In contrast to FARS, which applies structured generation procedures without continuous verification feedback, the proposed framework dynamically updates the contract state using evaluation signals to guide subsequent generation.
- We present an empirical evaluation with expert reviewers across multiple language model backbones and manuscript scenarios, demonstrating improved structural integrity, visual consistency, and overall manuscript robustness compared with DirectChat and FARS. These results suggest that contract-governed generation improves the reliability of automated scientific writing systems.

2. Related Work

2.1. Modular Approaches to Complex Text Generation

Recent work [40, 41, 42] has explored decomposing complex generation tasks into modular subtasks to improve controllability and output quality. Decomposed Prompting breaks down complex tasks into simpler subproblems that can be solved independently and then composed, demonstrating improved performance on multi-step reasoning tasks through explicit task decomposition [43, 44, 45, 46, 47]. Take a Step Back proposes evoking reasoning via abstraction by first generating high-level principles before tackling specific instances, showing that intermediate abstraction layers can guide more coherent problem-solving. In the visual domain, Training-Free Structured Diffusion introduces compositional guidance mechanisms that maintain structural constraints during image generation through explicit intermediate representations [11, 48, 49]. While these approaches demonstrate the value of intermediate representations and modular decomposition, they operate within single-agent frameworks that lack specialized expertise for different subtask types. Furthermore, these methods do not maintain bidirectional provenance tracking between intermediate representations and final outputs, making it difficult to trace which parts of the generated content derive from specific intermediate decisions [50, 51, 52]. However, these modular approaches do not address the challenge of maintaining global consistency across multiple interdependent sections in long-form documents, which our method resolves by introducing structured research stories as a semantic intermediate layer with explicit provenance tracking between story fields and generated paper sections.

2.2. Multi-Agent Architectures for Structured Tasks

Multi-agent systems [53, 54, 55, 32, 33] have emerged as a paradigm for tackling complex tasks through special-

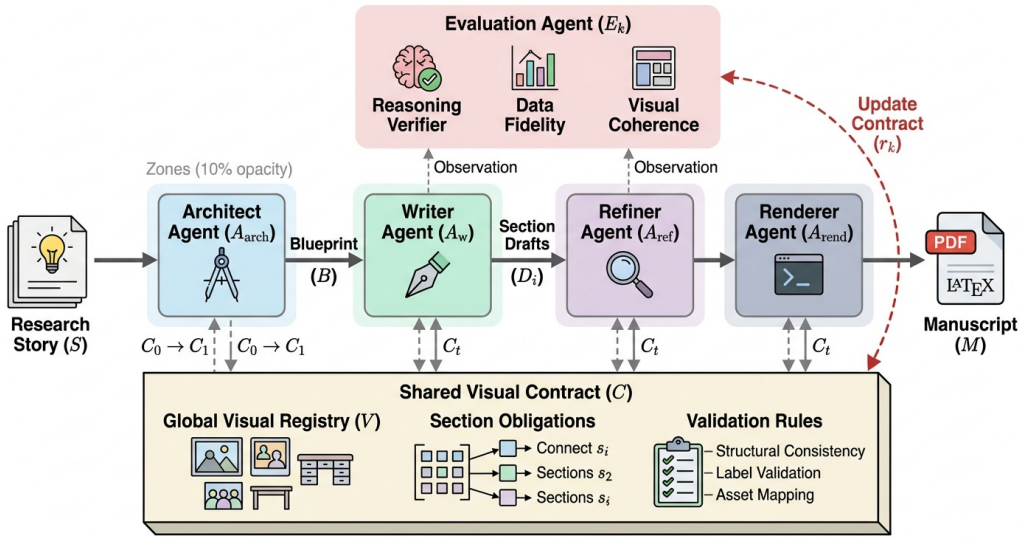

Figure 1. System overview of Story2Proposal. Given a research story as input, the framework coordinates architect, writer, refiner, and renderer agents through a persistent shared visual contract that records section structure, registered visual artifacts, and validation rules. Evaluation agents inspect intermediate outputs and feed corrective signals back to the contract state, enabling provenance-aware planning, structurally constrained drafting, global refinement, and deterministic rendering of the final manuscript.

ized agent collaboration. AgentSquare [43] proposes automatic agent search in modular design spaces, demonstrating that composing specialized agents with distinct capabilities can outperform monolithic models on complex reasoning tasks [56, 57]. Socratic Models introduces a framework for composing zero-shot multimodal reasoning by orchestrating multiple language models, each specialized for different modalities, through structured dialogue protocols [58]. Gamma Sampling provides fine-grained control over language model outputs without training by dynamically adjusting sampling distributions based on constraint satisfaction, enabling controllable generation through inference-time intervention [35, 59]. Confronting Reward Model Overoptimization addresses quality control in language model outputs through constrained reinforcement learning, establishing mechanisms for maintaining output quality during iterative refinement [60, 61]. These multi-agent and controllable generation approaches demonstrate the benefits of specialization and coordination, yet they lack grounding in structured intermediate semantic representations that explicitly encode domain-specific knowledge [62]. Moreover, existing multi-agent systems for text generation do not maintain revision memory across iterative refinement cycles, leading to inconsistent modifications when addressing critique feedback [63, 64]. However, these systems do not provide provenance-grounded coordination mechanisms that trace generated content back to specific semantic fields in structured input representations, which our approach addresses through discourse planning that maps story fields to section-level generation tasks and maintains cross-section consistency through shared claim-evidence maps.

3. Method

Story2Proposal models automated scientific manuscript generation as a coordinated multi-agent ecosystem governed by a shared structural contract, reflecting recent perspectives that frame large language models as cooperative agents capable of coordinating specialized functions to accomplish complex tasks [65]. Rather than treating generation as a single-pass language modeling task, the framework decomposes manuscript creation into specialized agents that operate over a persistent representation of structural, visual, and citation obligations. [66] Four generation agents—architect, writer, refiner, and renderer—interact with evaluation agents that assess reasoning quality, data fidelity, and visual consistency while generation proceeds. These agents communicate through an evolving contract state that constrains generation and adapts in response to evaluation feedback. The overall architecture is illustrated in Figure 1.

This design follows the paradigm of role-specialized collaborative language agents, which improve complex task decomposition and reliability compared with monolithic generation systems [32]. Multi-agent ecosystems enable structured coordination in which different agents focus on planning, generation, and verification stages of reasoning [67]. Story2Proposal extends this paradigm by introducing a persistent shared visual contract that enforces document-level structural and visual constraints throughout the generation pipeline.

3.1. Problem Formulation

Let S denote an input research story consisting of narrative descriptions, experimental evidence, and contextual information describing a scientific contribution. The objective is to transform S into a structured manuscript artifact M suitable for academic publication.

Unlike conventional generation pipelines that treat a manuscript as an unconstrained text sequence, Story2Proposal introduces an explicit contract governing structural and visual requirements. The contract specifies section organization, required visual artifacts, and reference relationships that must be satisfied during generation.

Let C denote the shared visual contract and V the set of required visual elements such as figures and tables. Let $\{s_i\}$ denote the ordered set of manuscript sections. The contract specifies how elements of V must appear within sections s_i and how they are referenced.

Manuscript construction proceeds through a sequence of contract-governed transformations:

$$M = A_{rend}(A_{ref}(A_w(A_{arch}(S, C_0), C_1), C_2), C_3) \quad (1)$$

where A_{arch}, A_w, A_{ref}, and A_{rend} represent the architect agent, writer agent, refiner agent, and renderer agent respectively. The contract evolves across stages as $\{C_t\}$.

Evaluation agents provide feedback during generation. Given evaluation signals $\{r_k\}$ from evaluation agents, the contract state is updated as

$$C_{t+1} = \text{Update}(C_t, \{r_k\}) \quad (2)$$

The final manuscript therefore emerges from coordinated agent interactions constrained by a continuously updated contract rather than a static prompt.

3.2. Notation

We define the key symbols used throughout the method in Table 1. The notation describes the input research story, the shared visual contract and its state transitions, the generation agents, the evaluation agents, and the produced manuscript artifact.

Symbol	Meaning
S	input research story and supporting evidence
C	shared visual contract encoding global and section-level visual obligations
C_t	state of the visual contract at generation stage t
A_{arch}	architect agent generating the manuscript blueprint
A_w	writer agent producing section drafts under contract constraints
A_{ref}	refiner agent performing global coherence alignment and compression
A_{rend}	renderer agent producing final LaTeX with deterministic validation
E_k	evaluation agent k assessing scientific reasoning, data fidelity, or visual coherence
M	generated manuscript artifact
V	set of required visual elements (figures and tables)
s_i	manuscript section i
r_k	evaluation feedback signal produced by evaluation agent k
Δ	performance delta between compared generation methods

Table 1. Notation used throughout the method.

3.3. Shared Visual Contract

The shared visual contract C provides the structural mechanism that coordinates generation across agents. It acts as a persistent representation that records all visual and structural obligations associated with the manuscript.

Figure 2 illustrates the contract schema.

The contract contains three layers of information.

First, a global visual registry maintains the set V of required visual artifacts. Each entry records the artifact type (figure or table), semantic description, canonical label identifier, and expected reference locations within the manuscript.

Second, section-level obligations specify which visual elements must appear within each section s_i. These constraints ensure that narrative explanations remain aligned with the figures or tables supporting the scientific claims.

Third, validation rules enforce document-wide consistency requirements such as unique labels, valid cross-references, and alignment between visual descriptions and their textual context.

The contract design is inspired by structured guardrail mechanisms that constrain language model outputs through explicit rule systems and taxonomies [68]. In contrast to safety guardrails, however, the contract in Story2Proposal enforces scientific document structure and visual integrity.

Embedding the contract as a persistent shared state allows

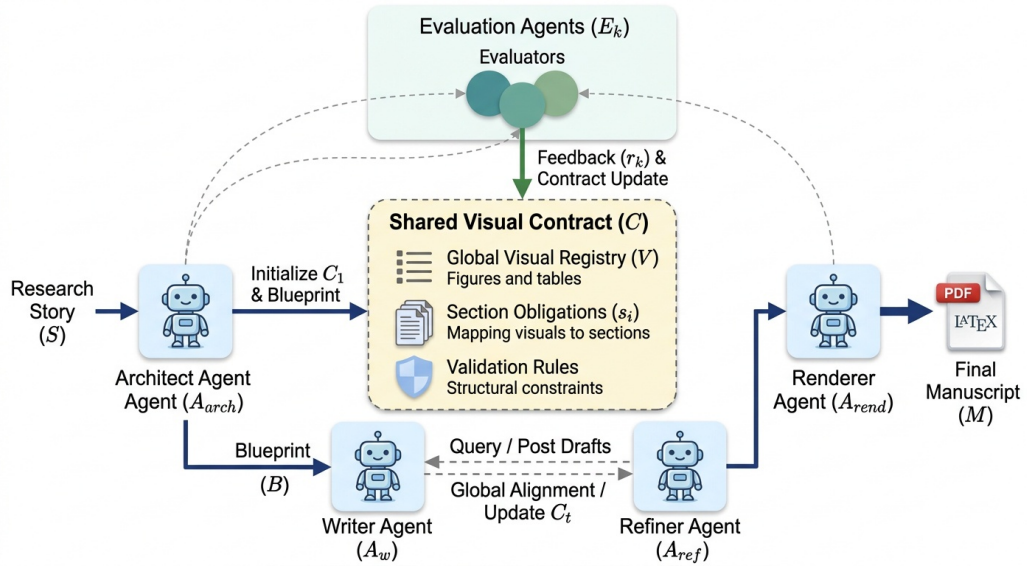

Figure 2. Schema of the shared visual contract used by Story2Proposal. The contract combines a global registry of figures, tables, and citation slots with section-level obligations and validation rules, so that all agents operate on the same structural state and the renderer can deterministically verify label uniqueness, reference resolution, and narrative–visual alignment before compilation.

agents to remain aware of structural obligations during generation. This prevents missing figures, misplaced references, or inconsistent visual descriptions that commonly arise in unconstrained text generation.

3.4. Multi-Agent Generation Pipeline

Manuscript generation proceeds through a coordinated pipeline of four agents operating over the shared contract state. The pipeline is illustrated in Figure **??**.

The architect agent transforms the research story S into a structured manuscript blueprint while initializing the contract. It decomposes the narrative into an ordered section structure $\{s_i\}$ and specifies the argument outline for each section, including key claims and supporting evidence. The architect also identifies candidate visual artifacts and registers them in the contract's visual registry V, assigning semantic descriptions and canonical labels. Each artifact is then mapped to section-level obligations indicating where it must appear in the manuscript. Formally, the architect produces a blueprint B and updated contract state:

$$(B, C_1) = A_{arch}(S, C_0) \tag{3}$$

The writer agent generates section drafts that realize the blueprint while satisfying contract constraints. Given a section specification s_i and contract state C_t, the writer produces a draft D_i that follows the planned argument structure and includes the required visual references registered in the contract. Visual markers and citation identifiers must match the contract registry to maintain consistent references throughout the document. The drafting step can be expressed as

$$D_i = A_w(s_i, C_t) \tag{4}$$

The refiner agent performs global alignment over the set of generated drafts $\{D_i\}$. Its role is to improve coherence and consistency across sections while reconciling the manuscript with the contract state. The refiner compresses redundant explanations, harmonizes terminology across sections, and ensures that each visual element referenced in the manuscript is described appropriately in its surrounding text. If inconsistencies are detected, the refiner may trigger contract updates through evaluation feedback. The refinement stage produces a consolidated manuscript:

$$(M', C_{t+1}) = A_{ref}(\{D_i\}, C_t) \tag{5}$$

The renderer agent converts the refined manuscript into stable LaTeX output while enforcing deterministic structural validation. During this stage, all visual references are resolved, label identifiers are standardized, and cross-references are validated against the contract. Each artifact in V must appear exactly once and must be referenced consistently within the text. The renderer outputs the final manuscript artifact:

$$M = A_{rend}(M', C_t) \tag{6}$$

Separating planning, drafting, refinement, and rendering

reduces error propagation and allows each agent to operate with specialized context. The architect establishes the structural skeleton, the writer produces local drafts, the refiner enforces global narrative consistency, and the renderer guarantees structural correctness at the document level.

3.5. Evaluation and Contract Updates

Evaluation agents monitor intermediate artifacts during generation and provide feedback signals used to update the contract state. Let $\{E_k\}$ denote the set of evaluation agents, each responsible for a specific dimension such as reasoning verification, data fidelity assessment, or visual consistency.

Given an intermediate artifact X and contract state C_t, evaluation agent E_k produces a feedback signal

$$r_k = E_k(X, C_t) \tag{7}$$

These signals describe detected issues, confidence estimates, or recommended corrections. Feedback signals are aggregated to update the contract:

$$C_{t+1} = \text{Update}(C_t, \{r_k\}) \tag{8}$$

Contract updates may introduce additional validation rules, modify visual placement constraints, or require additional explanatory context for specific artifacts. For example, if an evaluation agent detects that a figure reference lacks supporting explanation, the contract may require a descriptive paragraph to accompany the reference.

Embedding evaluation within the generation pipeline enables early detection of inconsistencies and prevents structural errors from propagating through later stages. This generate–evaluate–adapt mechanism reflects emerging architectures for autonomous agent ecosystems, where agents coordinate through shared environments and communication protocols to solve complex tasks collaboratively [69]. Such ecosystems support adaptive reasoning and coordinated decision-making across specialized agents [70].

3.6. Optimization Objective and Algorithm

Although Story2Proposal does not train a single monolithic model, the evaluation feedback signals can be interpreted as rewards guiding system-level optimization. Let $R(M)$ denote the aggregated evaluation score for a generated manuscript M:

$$R(M) = \sum_k w_k r_k \tag{9}$$

where r_k represents the feedback from evaluation agent E_k and w_k denotes the relative importance assigned to each evaluation dimension. The generation objective is to maximize $R(M)$ while satisfying all constraints encoded in the contract.

The overall procedure can be summarized as follows:

Algorithm 1 Story2Proposal Generation Procedure

Input: research story S, initial contract C_0, evaluators
Output: manuscript M
$(B, C) \leftarrow architect(S, C_0)$
drafts $\leftarrow [\,]$
for each section in $B.sections$ do
  $D \leftarrow writer(section, C)$
  feedback $\leftarrow [E(D, C)\ for\ E\ in\ evaluators]$
  $C \leftarrow update_contract(C, feedback)$
  append D to drafts
end for
$(M_prime, C) \leftarrow refiner(drafts, C)$
feedback $\leftarrow [E(M_prime, C)\ for\ E\ in\ evaluators]$
$C \leftarrow update_contract(C, feedback)$
$M \leftarrow renderer(M_prime, C)$
return M

The algorithm highlights the core principle of Story2Proposal: manuscript generation emerges from coordinated interactions among specialized agents operating under a shared contract that evolves through evaluation feedback. This mechanism transforms the writing process from a static generation pipeline into an adaptive ecosystem where generation and verification continuously guide one another.

4. Experiments

4.1. Implementation Details

The Story2Proposal system was implemented as a coordinated pipeline of four agents (architect agent, writer agent, refiner agent, and renderer agent) that composed the generation workflow. [71] Each agent operated as a language-model-driven component interacting through the shared visual contract and the evolving contract state. Agents executed sequentially within the generate–evaluate–adapt loop described in Section 3, where evaluation agents analyzed intermediate artifacts and produced feedback signals that updated the contract state before subsequent generation stages, reflecting emerging practices in evaluating autonomous agent behavior in realistic task environments [72]. To evaluate robustness across model backbones, the same pipeline configuration was executed using four widely used large language models: GPT, Claude, Gemini, and Qwen. Each run generated a complete manuscript artifact from the same research story input while enforcing contract-governed structural constraints and validating visual artifacts during rendering. The renderer agent produced

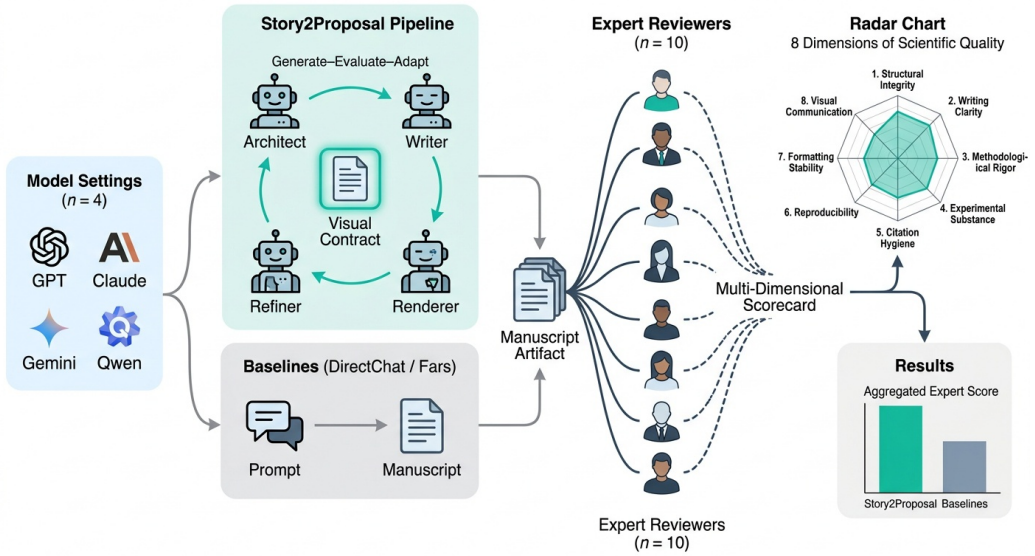

Figure 3. Expert evaluation protocol used in our experiments. Each condition combines one of two generation methods, one of four LLM backbones (GPT, Claude, Gemini, and Qwen), and ten independent expert reviewers. Reviewers assess complete manuscripts on structural integrity, writing clarity, methodological rigor, experimental substance, citation hygiene, reproducibility, formatting stability, and visual communication, and the resulting scores are aggregated into the overall expert evaluation metric.

deterministic LaTeX output and verified compliance with the persistent contract before finalizing the manuscript artifact used for evaluation, and all compared systems were executed with identical prompts and experimental inputs to ensure fairness across model settings and baseline methods in a controlled multi-agent evaluation setup [73].

4.2. Experimental Design

The evaluation protocol assessed whether contract-governed generation improved the robustness and structural reliability of automatically generated research manuscripts. [74] The benchmark environment used the Jericho research corpus to construct research stories and manuscript tasks, enabling evaluation of complex reasoning and structured document synthesis in a controlled setting. Human expert reviewers served as the primary evaluation mechanism because automated metrics often fail to capture reasoning coherence and document-level scientific quality in generated manuscripts [75]. Prior work has similarly emphasized the importance of human-grounded evaluation when assessing complex agent systems and generation pipelines [76]. The evaluation followed a $2 \times 4 \times 10$ design consisting of two generation methods, four model settings, and ten independent expert reviewers per condition. Reviewers evaluated manuscripts across eight dimensions of scientific writing quality: structural integrity, writing clarity, methodological rigor, experimental substance, citation hygiene, reproducibility, formatting stability, and visual communication. Scores across these dimensions were aggregated into a single expert evaluation score for each generated manuscript. Figure 3 illustrates the full evaluation design including model configurations and reviewer structure. The first experiment compared Story2Proposal with DirectChat, a single-stage prompt-based manuscript generation baseline without structural contracts or multi-agent coordination. This comparison tested whether contract-governed generation improved document-level organization and reasoning coherence relative to direct language model generation. The second experiment compared Story2Proposal with Fars, a structured manuscript generation baseline evaluated on four benchmark research papers. This comparison examined whether the contract-centered architecture provided improvements even against a structured system designed for research manuscript synthesis. The protocol follows established practices for evaluating complex multi-agent workflows using structured output analysis and systematic scoring procedures [77].

4.3. Results

Table 2 reports the primary comparison between Story2Proposal and DirectChat across four model settings. Story2Proposal outperformed DirectChat in every configuration, indicating that contract-governed generation improved manuscript quality independently of the underlying language model. With GPT, Story2Proposal achieved an expert evaluation score of 5.962 compared with 4.078 for DirectChat. Claude produced 6.153 versus 3.864, Gemini produced 6.257 versus 3.934, and Qwen produced 6.207 versus 3.975. Averaged across all model settings, Story2Proposal achieved 6.145 compared with 3.963 for DirectChat, an

Table 2. Average expert evaluation scores comparing Story2Proposal and DirectChat across four LLM settings. Higher is better.

Model	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	Avg
Based on DirectChat											
GPT	3.62	3.01	5.97	3.33	3.49	4.06	5.34	4.23	3.64	4.09	4.08
Claude	3.39	2.68	5.98	3.04	3.36	3.96	4.88	3.92	3.23	4.20	3.86
Gemini	3.32	2.79	6.07	3.16	3.51	3.66	5.14	3.99	3.56	4.14	3.93
Qwen	3.38	2.74	6.19	3.22	3.41	3.93	5.32	4.05	3.58	3.93	3.98
Based on Story2Proposal											
GPT	5.39	4.99	6.98	5.03	5.18	6.25	7.13	7.60	5.09	5.98	5.96
Claude	5.70	5.53	7.22	5.35	5.30	6.01	6.83	7.81	5.58	6.20	6.15
Gemini	5.60	5.61	7.42	5.51	5.59	6.17	7.23	7.54	5.52	6.38	6.26
Qwen	6.07	5.60	7.24	5.35	5.44	6.13	7.09	7.20	5.83	6.12	6.21

improvement of +2.182. These results indicated that the contract-governed multi-agent framework consistently produced higher-quality manuscripts across diverse model backbones. Expert reviewers reported fewer structural inconsistencies, clearer experimental descriptions, and more reliable formatting in Story2Proposal outputs. Improvements were particularly visible in structural integrity and formatting stability, where manuscripts generated through the contract maintained more consistent section organization and fewer layout errors than those produced by DirectChat.

Table 3 presents the benchmark comparison between Story2Proposal and Fars across four research papers: Escrowed Batch Reveal, Symbolic Execution, Hazard-Signature Tombstones, and Poisoning LLM-Induced Rules. Story2Proposal achieved higher expert evaluation scores on all four papers. For Escrowed Batch Reveal, the score increased from 4.928 with Fars to 5.459 with Story2Proposal. For Symbolic Execution, the score increased from 5.638 to 5.886. Hazard-Signature Tombstones improved from 4.709 to 5.600, and Poisoning LLM-Induced Rules improved from 5.511 to 5.876. Averaged across the four benchmark papers, Story2Proposal achieved a mean expert evaluation score of 5.705 compared with 5.197 for Fars, corresponding to an improvement of +0.508. Although the margin was smaller than in the DirectChat comparison, the results indicated that maintaining an evolving contract state and enforcing structural obligations during generation improved document reliability even relative to a structured baseline. Expert reviewers also reported stronger alignment between narrative explanations and visual artifacts, particularly in experimental sections where figures and tables were consistently referenced and correctly positioned. No ablation study was included in this version of the system evaluation.

5. Analysis

5.1. Structural Robustness and Feedback Mechanisms

The primary improvement observed for Story2Proposal arose from enforcing structural constraints through the persistent shared visual contract during manuscript generation,

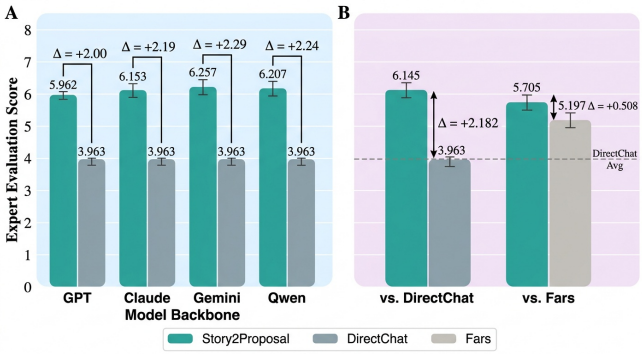

Figure 4. Average performance deltas between Story2Proposal and baseline generation methods across experiments. Positive values indicate that the contract-governed multi-agent framework outperforms the comparison system under the same backbone or benchmark setting. The figure summarizes both the large gains over DirectChat in the cross-model study and the smaller but consistent improvements over the structured Fars baseline, highlighting the benefit of stronger structural control and narrative–visual alignment.

a strategy consistent with prior work showing that explicit planning and structured narrative scaffolds can improve coherence and organization in generated outputs [78]. As illustrated in Figure 4, manuscripts produced by Story2Proposal consistently achieved higher expert evaluation scores than those generated by DirectChat across all evaluated model backbones. The system reached an average expert evaluation score of 6.145 compared with 3.963 for DirectChat, corresponding to an improvement of +2.182. Because this improvement appeared consistently across different model backbones, the results indicate that the advantage derived primarily from the contract-governed architecture rather than from any specific language model capability, aligning with broader evidence that structured multi-component or collaborative reasoning systems can outperform single-agent generation pipelines on complex tasks [79].

A central mechanism behind this improvement is the contract state maintained throughout the generation process. The persistent shared visual contract specifies section structure, visual artifacts, and validation rules that must be satisfied during manuscript construction. By embedding these

Table 3. Average expert evaluation scores comparing Story2Proposal and Fars across four benchmark research papers. Higher is better.

Paper	E1	E2	E3	E4	E5	E6	E7	E8	E9	E10	Avg
Based on Fars											
Escrowed Batch Reveal	4.55	4.39	6.06	4.41	4.38	5.04	6.19	4.57	4.47	5.22	4.93
Symbolic Execution	4.97	5.64	6.32	5.40	5.65	5.57	6.46	5.11	5.57	5.69	5.64
Hazard-Signature Tombstones	4.27	4.06	6.09	4.19	4.33	4.57	6.04	4.59	4.21	4.74	4.71
Poisoning LLM-Induced Rules	5.00	5.59	6.01	5.11	5.35	5.22	6.80	5.26	5.14	5.63	5.41
Based on Story2Proposal											
Escrowed Batch Reveal	5.72	4.49	6.61	4.62	4.67	5.27	6.94	5.87	4.87	5.53	5.46
Symbolic Execution	5.66	5.20	6.79	5.16	5.34	5.67	7.54	5.90	5.28	6.32	5.89
Hazard-Signature Tombstones	4.96	4.87	6.52	4.74	5.24	5.38	7.00	6.00	5.29	6.00	5.60
Poisoning LLM-Induced Rules	5.61	5.22	6.68	4.79	5.56	5.88	7.15	6.38	5.32	6.17	5.88

constraints directly into generation, Story2Proposal prevents common failure modes of unconstrained generation pipelines such as structural drift, missing visual artifacts, or inconsistent section organization. [80] Agents therefore operate within an explicit structural specification rather than relying on implicit formatting decisions during text generation.

This behavior aligns with prior work showing that structured constraints can improve reliability in large language model generation. Grammar-guided generation systems demonstrate that restricting output structure enables earlier detection and correction of inconsistencies during generation rather than after completion [81]. Story2Proposal applies a similar principle at the document level: each stage of the pipeline updates and validates the contract state before subsequent generation steps proceed, preventing structural errors from accumulating across sections.

An additional contributor to robustness is the generate–evaluate–adapt loop that integrates evaluation agents into the generation process. Instead of performing evaluation only after the manuscript is produced, intermediate artifacts are continuously assessed for reasoning quality, structural integrity, and visual consistency. Feedback signals from evaluation agents modify the contract state, enabling the system to correct issues early in the pipeline.

This feedback-driven process resembles interpretable reasoning frameworks that expose intermediate reasoning stages so that errors can be detected before final outputs are produced. Such approaches improve controllability and transparency by making intermediate reasoning states observable and correctable [82]. In Story2Proposal, evaluation agents play an analogous role by monitoring intermediate manuscript states and triggering corrective updates when inconsistencies are detected.

5.2. Cross-Model Consistency and Visual Communication Effects

A key empirical observation is that the performance improvement remained stable across multiple language model backbones. Story2Proposal consistently outperformed DirectChat for GPT, Claude, Gemini, and Qwen. The system achieved expert evaluation scores of 5.962 for GPT, 6.153 for Claude, 6.257 for Gemini, and 6.207 for Qwen, while DirectChat produced substantially lower scores across the same models. This pattern suggests that the architectural structure of Story2Proposal, rather than model-specific reasoning ability, primarily explains the observed improvements.

In unconstrained generation pipelines such as DirectChat, the language model must simultaneously manage reasoning, formatting, section structure, and visual references. This increases the complexity of the generation task and often leads to structural inconsistencies. Story2Proposal externalizes these responsibilities through the persistent shared visual contract and specialized agents, allowing the language model to focus on localized reasoning and content generation while structural constraints are handled by the system architecture.

The benefits of such decomposition are consistent with findings from multi-agent simulation environments where distributing responsibilities across coordinated agents improves reliability and stability in complex workflows [83]. By separating planning, drafting, refinement, and rendering responsibilities, Story2Proposal maintains consistent structural behavior even when the underlying language model changes.

The visual contract also directly improved visual communication and formatting stability. In prompt-based generation pipelines, figures and tables are often lost or duplicated during iterative rewriting because they are represented only as text tokens. Story2Proposal instead registers visual artifacts in a contract-based registry with explicit identifiers and placement obligations. Agents reference these registered artifacts rather than regenerating figure markers, ensuring that visual elements remain aligned with the narrative throughout the document.

This design difference is reflected in the comparison with the Fars baseline. Across the four benchmark papers—Escrowed Batch Reveal, Symbolic Execution, Hazard-

Signature Tombstones, and Poisoning LLM-Induced Rules—Story2Proposal achieved higher expert evaluation scores than Fars in every case, with an average score of 5.705 compared with 5.197 for Fars, corresponding to an improvement of +0.508. Because Fars already employs structured templates, the remaining improvement suggests that persistent contract enforcement and iterative validation further strengthen alignment between narrative explanations and visual artifacts.

Prior studies on visual communication systems also highlight that maintaining explicit alignment between narrative text and visual transformations improves interpretability in complex documents [84]. By encoding such alignment constraints directly within the contract state, Story2Proposal preserves the semantic relationship between visual artifacts and their textual explanations during generation.

5.3. Failure Mode Analysis

Despite the consistent gains observed in the experiments, the results also reveal limitations of the contract-governed generation approach. The improvement over the Fars baseline was smaller than the improvement over DirectChat. While Story2Proposal outperformed Fars across all evaluated benchmark papers, the margin of improvement was notably reduced. This pattern indicates that the largest benefits arise when replacing unconstrained prompt-based generation systems, whereas structured baselines already mitigate many basic structural errors.

Another limitation arises from the reliance on evaluation agents for detecting inconsistencies. If evaluation agents fail to identify subtle reasoning weaknesses—such as an interpretation that satisfies structural requirements but remains scientifically weak—the contract state may not trigger corrective updates. In such cases, the generated manuscript can remain structurally valid while containing less rigorous reasoning.

Improving this stage may require richer evaluation signals and more specialized critique modules. Structured risk taxonomies and evaluation frameworks have been shown to improve the identification of nuanced categories in language model outputs [68], suggesting that more sophisticated evaluation agents could strengthen the feedback mechanism in future systems.

A final limitation concerns incomplete research inputs. Because Story2Proposal enforces structural and visual obligations rather than generating missing scientific evidence, manuscripts derived from incomplete research stories may remain logically consistent yet still reflect gaps in the underlying narrative. Addressing this issue would likely require integrating retrieval mechanisms or evidence verification systems capable of expanding or validating the input research story before manuscript generation begins.

6. Conclusion

This work presented Story2Proposal, a contract-governed multi-agent framework that models automated scientific manuscript generation as a structured construction process in which specialized agents transform a research story into a coherent manuscript while enforcing shared structural and visual obligations. Empirical results showed consistent improvements over unconstrained generation pipelines: across four model backbones, Story2Proposal achieved expert evaluation scores of 5.962 (GPT), 6.153 (Claude), 6.257 (Gemini), and 6.207 (Qwen), while DirectChat produced 4.078, 3.864, 3.934, and 3.975, yielding an overall score of 6.145 versus 3.963 and an improvement of +2.182 points. Comparisons with the structured baseline Fars also demonstrated gains in document reliability and alignment between narrative text and visual artifacts, with Story2Proposal averaging 5.705 versus 5.197, including improvements on Escrowed Batch Reveal (5.459 vs 4.928), Symbolic Execution (5.886 vs 5.638), Hazard-Signature Tombstones (5.600 vs 4.709), and Poisoning LLM-Induced Rules (5.876 vs 5.511). These findings indicate that maintaining a persistent shared visual contract and evolving contract state helps preserve structural integrity, formatting stability, and visual communication during generation, suggesting that architectural constraints play a central role in reliable manuscript construction. Improvements over template-based systems such as Fars remain smaller, and performance still depends on the completeness of the input research story and the ability of evaluation agents to detect subtle reasoning issues. Future work may extend contract-centered generation with stronger verification and domain-specific validation mechanisms for reliable AI-assisted scientific communication while supporting interpretable reasoning in automated analysis systems.

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